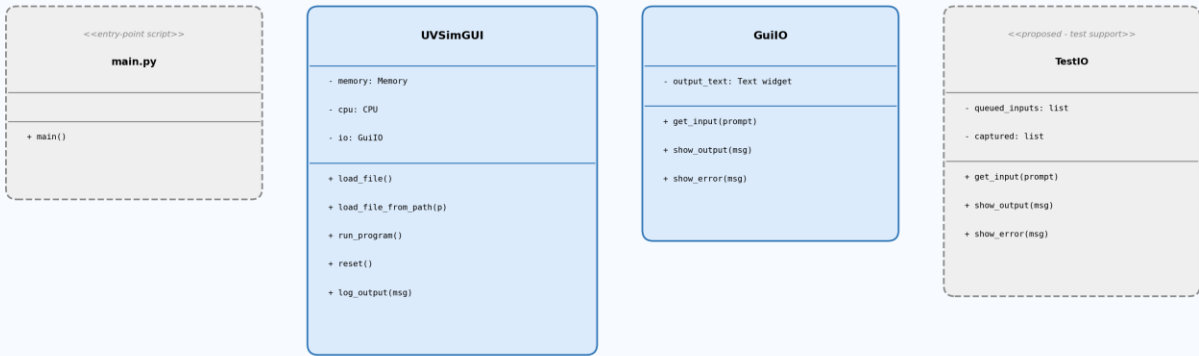
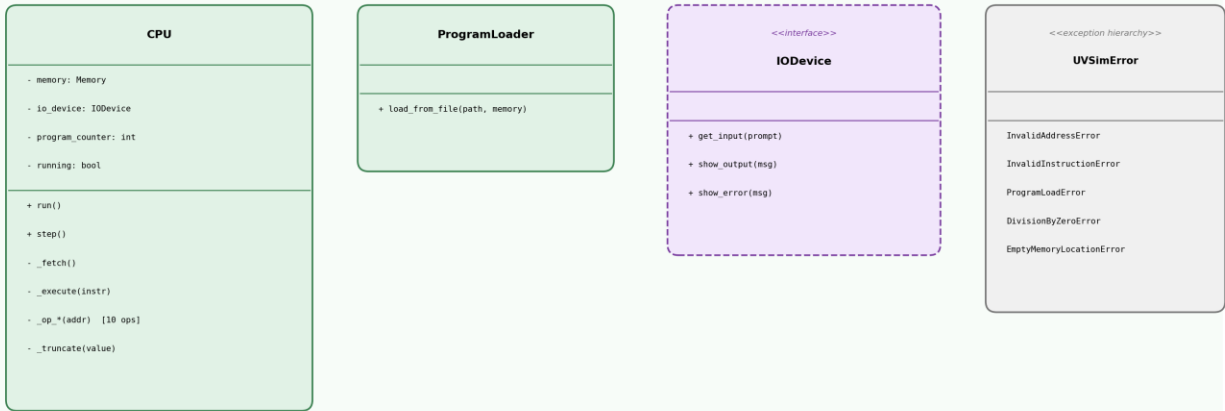


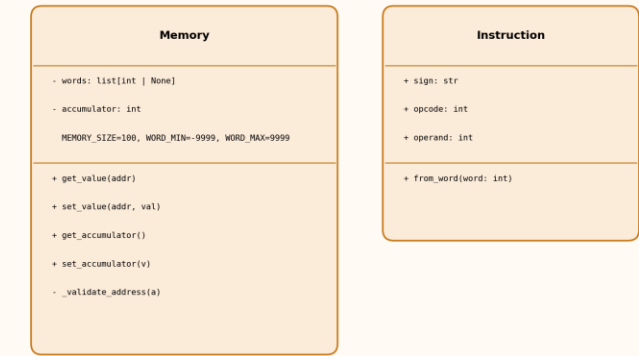
PRESENTATION LAYER



BUSINESS LOGIC LAYER



DATA ACCESS LAYER



Classes

Memory

Purpose: Stores the simulator's memory words and accumulator value. Provides safe access to memory locations.

Instruction

Purpose: Represents a machine-language instruction and separates it into opcode and operand components.

CPU

Purpose: Executes instructions, manages program flow, and coordinates communication between memory and I/O devices.

IODevice

Purpose: Defines a common interface for input and output operations.

GuiIO

Purpose: Implements the I/O interface for the graphical user interface.

TestIO

Purpose: Implements the I/O interface for automated testing.

ProgramLoader

Purpose: Loads machine-language programs into memory from external files.

UVSimError

Purpose: Base exception class for simulator-specific errors.

InvalidAddressError

Purpose: Raised when an invalid memory address is accessed.

InvalidInstructionError

Purpose: Raised when an invalid instruction is encountered.

ProgramLoaderError

Purpose: Raised when a program cannot be loaded correctly.

DivisionByZeroError

Purpose: Raised when a division-by-zero operation occurs.

EmptyMemoryLocationError

Purpose: Raised when an uninitialized memory location is accessed.

Functions

Memory

read(address)

Purpose: Returns the value stored at a memory address.

write(address, value)

Purpose: Stores a value at a memory address.

get_accumulator()

Purpose: Returns the current accumulator value.

set_accumulator(value)

Purpose: Updates the accumulator value.

_validate_address(address)

Purpose: Verifies that an address is within valid memory bounds.

Instruction

from_word(word)

Purpose: Converts a machine word into an Instruction object.

CPU

run()

Purpose: Executes instructions until a halt instruction is reached.

step()

Purpose: Executes a single instruction.

_fetch()

Purpose: Retrieves the next instruction from memory.

_execute(instruction)

Purpose: Performs the operation specified by an instruction.

IODevice

read()

Purpose: Obtains input from a data source.

write(value)

Purpose: Outputs a value to a data destination.

GuiO

read()

Purpose: Receives input from the GUI.

write(value)

Purpose: Displays output in the GUI.

TestIO

read()

Purpose: Provides predefined input values for testing.

write(value)

Purpose: Records output values for test verification.

ProgramLoader

load_from_file(path, memory)

Purpose: Reads a program file and loads its contents into memory.